Question–1 [JAM–2005]

Let E, F and G be three events such that the events E and F are mutually exclusive, $P(E \cup F) = 1$, $P(E \cap G) = \frac{1}{4}$ and $P(G) = \frac{7}{12}$. Then $P(F \cap G)$ equals

(A) $\frac{1}{12}$

(B) $\frac{1}{4}$

(C) $\frac{5}{12}$

(D) $\frac{1}{3}$

Solution

Given:

$$\begin{aligned} E \cap F &= \emptyset \quad \text{(since } E \text{ and } F \text{ are mutually exclusive)} \\ P(E \cup F) &= 1, \\ P(E \cap G) &= \frac{1}{4}, \\ P(G) &= \frac{7}{12}. \end{aligned}$$

Since $P(E \cup F) = 1$, we have

$$E \cup F = S,$$

where S denotes the sample space.

Now,

$$G = G \cap S.$$

Since $S = E \cup F$,

$$G = G \cap (E \cup F).$$

Using the distributive law,

$$G = (E \cap G) \cup (F \cap G).$$

Because E and F are mutually exclusive,

$$(E \cap G) \cap (F \cap G) = \emptyset.$$

Hence,

$$P(G) = P(E \cap G) + P(F \cap G).$$

Substituting the given values,

$$\frac{7}{12} = \frac{1}{4} + P(F \cap G).$$

Therefore,

$$P(F \cap G) = \frac{7}{12} - \frac{1}{4}.$$

Taking the LCM as 12,

$$P(F \cap G) = \frac{7}{12} - \frac{3}{12} = \frac{4}{12} = \frac{1}{3}.$$

$$\boxed{P(F \cap G) = \frac{1}{3}}$$

Question–2 [JAM–2005]

There are four urns labeled U_1, U_2, U_3 and U_4, each containing 3 blue and 5 red balls. The fifth urn, labeled U_5, contains 4 blue and 4 red balls. An urn is selected at random from these five urns and a ball is drawn at random from it. Given that the selected ball is red, find the probability that it came from the urn U_5

Solution

Let

$$R = \text{“The selected ball is red.”}$$

Since each urn is selected with equal probability,

$$P(U_i) = \frac{1}{5}, \qquad i = 1, 2, \ldots, 5.$$

For the first four urns,

$$P(R|U_i) = \frac{5}{8}, \qquad i = 1, 2, 3, 4.$$

For the fifth urn,

$$P(R|U_5) = \frac{4}{8} = \frac{1}{2}.$$

Using the Law of Total Probability,

$$\begin{aligned} P(R) &= \sum_{i=1}^{5} P(R|U_i)P(U_i) \\ &= 4\left(\frac{5}{8} \cdot \frac{1}{5}\right) + \frac{1}{2} \cdot \frac{1}{5} \\ &= \frac{1}{2} + \frac{1}{10} \\ &= \frac{3}{5}. \end{aligned}$$

By Bayes' theorem,

$$\begin{aligned} P(U_5|R) &= \frac{P(R|U_5)P(U_5)}{P(R)} \\ &= \frac{\frac{1}{2} \cdot \frac{1}{5}}{\frac{3}{5}} \\ &= \frac{1}{10} \times \frac{5}{3} \\ &= \boxed{\frac{1}{6}}. \end{aligned}$$

$$\boxed{P(U_5|R) = \frac{1}{6}.}$$

Question–3 [JAM–2005]

Let A, B and C be pair-wise independent events such that

$$P(A \cap B) = 0.1 \quad \text{and} \quad P(B \cap C) = 0.2.$$

Show that

$$P(A^c \cup C) \geq \frac{7}{8}.$$

Solution

Since the events are pairwise independent,

$$P(A)P(B) = P(A \cap B) = 0.1,$$

and

$$P(B)P(C) = P(B \cap C) = 0.2.$$

Hence,

$$P(A) = \frac{0.1}{P(B)}, \qquad P(C) = \frac{0.2}{P(B)}.$$

Now,

$$P(A^c \cup C) = 1 - P(A \cap C^c).$$

Also,

$$P(A \cap C^c) = P(A) - P(A \cap C).$$

Again, since A and C are pairwise independent,

$$P(A \cap C) = P(A)P(C).$$

Therefore,

$$P(A^c \cup C) = 1 - P(A) + P(A)P(C).$$

Substituting the expressions for $P(A)$ and $P(C)$,

$$P(A^c \cup C) = 1 - \frac{0.1}{P(B)} + \frac{0.02}{P(B)^2}.$$

Let

$$x = P(B), \qquad 0 < x \leq 1.$$

Define

$$f(x) = 1 - \frac{0.1}{x} + \frac{0.02}{x^2}.$$

Differentiating,

$$f'(x) = \frac{0.1}{x^2} - \frac{0.04}{x^3} = \frac{0.1x - 0.04}{x^3}.$$

Setting

$$f'(x) = 0$$

gives

$$x = 0.4.$$

Further,

$$f''(0.4) > 0,$$

so this gives the minimum value.
Hence,

$$\begin{aligned} P(A^c \cup C) &\geq 1 - \frac{0.1}{0.4} + \frac{0.02}{0.16} \\ &= 1 - \frac{1}{4} + \frac{1}{8} \\ &= \boxed{\frac{7}{8}}. \end{aligned}$$

Thus,

$$\boxed{P(A^c \cup C) \geq \frac{7}{8}.}$$

Question–4 [JAM–2006]

Let E and F be two mutually disjoint events. Further, let E and F be independent of G. If

$$p = P(E) + P(F) \quad \text{and} \quad q = P(G),$$

then

$$P(E \cup F \cup G)$$

is

(A) $1 - pq$

(B) $q + p^2$

(C) $p + q^2$

(D) $p + q - pq$

Solution

Given,

$$E \cap F = \emptyset,$$

and

$$P(E) + P(F) = p, \qquad P(G) = q.$$

Also,

$$E \text{ and } G$$

are independent, so

$$P(E \cap G) = P(E)P(G).$$

Similarly,

$$P(F \cap G) = P(F)P(G).$$

Using the Inclusion–Exclusion Principle,

$$P(E \cup F \cup G) = P(E) + P(F) + P(G) \\ - P(E \cap F) - P(E \cap G) - P(F \cap G) \\ + P(E \cap F \cap G).$$

Since

$$E \cap F = \emptyset,$$

we have

$$P(E \cap F) = 0, \qquad P(E \cap F \cap G) = 0.$$

Therefore,

$$P(E \cup F \cup G) = P(E) + P(F) + P(G) \\ - P(E)P(G) - P(F)P(G).$$

Substituting

$$P(E) + P(F) = p, \qquad P(G) = q,$$

we obtain

$$\boxed{P(E \cup F \cup G) = p + q - pq.}$$

Hence, the correct option is

$$\boxed{\text{(D) } p + q - pq.}$$

Question 5 (IIT JAM 2006)

An archer makes 10 independent attempts at a target and his probability of hitting the target at each attempt is $\frac{5}{6}$.

Then the conditional probability that his last two attempts are successful given that he has a total of 7 successful attempts is

$$(A)\ \frac{1}{5^5} \qquad (B)\ \frac{7}{15} \qquad (C)\ \frac{25}{36} \qquad (D)\ \frac{8!}{3!5!}\left(\frac{5}{6}\right)^7\left(\frac{1}{6}\right)^3$$

Concept Used

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Since the total number of successful attempts is already fixed (equal to 7), every arrangement of 7 successes among 10 positions is equally likely.

Therefore, probability can be obtained using combinations only.

Solution

Let

$$A = \{\text{last two attempts are successful}\}$$

$$B = \{\text{exactly 7 successful attempts}\}$$

We require

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

—

Total arrangements with exactly seven successes

$$\binom{10}{7} = 120.$$

Now fix the last two positions as successes.
Already 2 successes are fixed.
Hence among the first 8 positions we need

$$5 \text{ successes and } 3 \text{ failures.}$$

Number of favourable arrangements

$$\binom{8}{5} = 56.$$

Therefore

$$P(A|B) = \frac{\binom{8}{5}}{\binom{10}{7}} = \frac{56}{120} = \frac{7}{15}.$$

$$\boxed{\frac{7}{15}}$$

Hence the correct option is

$$\boxed{(B)}$$

Shortcut

Whenever the total number of successes is fixed, ignore the probability of success ($p = \frac{5}{6}$).

Use combinations only.

Question 6 (IIT JAM 2006)

Let A and B be two events with

$$P(A|B) = 0.3, \qquad P(A|B^c) = 0.4.$$

Find $P(B|A)$ and $P(B^c|A^c)$ in terms of $P(B)$.

If

$$\frac{1}{4} \leq P(B|A) \leq \frac{1}{3},$$

and

$$\frac{1}{4} \leq P(B^c|A^c) \leq \frac{9}{16},$$

determine $P(B)$.

Concept Used

1. Bayes' theorem
2. Law of Total Probability

Solution

Let

$$P(B) = p.$$

Then

$$P(B^c) = 1 - p.$$

Using total probability,

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c).$$

Hence

$$P(A) = 0.3p + 0.4(1 - p) = 0.4 - 0.1p.$$

Using Bayes theorem,

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)} = \frac{0.3p}{0.4 - 0.1p} = \frac{3p}{4 - p}.$$

Therefore

$$\boxed{P(B|A) = \frac{3P(B)}{4 - P(B)}}$$

—

Now

$$P(A^c|B) = 0.7,$$

$$P(A^c|B^c) = 0.6.$$

Hence

$$P(A^c) = 0.7p + 0.6(1 - p) = 0.6 + 0.1p.$$

Again by Bayes theorem,

$$P(B^c|A^c) = \frac{0.6(1 - p)}{0.6 + 0.1p} = \frac{6(1 - p)}{6 + p}.$$

Thus

$$\boxed{P(B^c|A^c) = \frac{6(1-P(B))}{6+P(B)}}$$

—

Using

$$\frac{1}{4} \leq \frac{3p}{4-p} \leq \frac{1}{3}$$

we obtain

$$\frac{2}{7} \leq p \leq \frac{2}{5}.$$

Using

$$\frac{1}{4} \leq \frac{6(1-p)}{6+p} \leq \frac{9}{16},$$

we obtain

$$p \leq \frac{2}{5}.$$

Combining both inequalities,

$$\boxed{\frac{2}{7} \leq P(B) \leq \frac{2}{5}}$$

Question 7 (IIT JAM 2007)

Independent trials consisting of rolling a fair die are performed.

The probability that 2 appears before 3 or 5 is

$$(A)\ \frac{1}{2} \qquad (B)\ \frac{1}{3} \qquad (C)\ \frac{1}{4} \qquad (D)\ \frac{1}{5}$$

Concept Used

Competing Events (First Occurrence Method)

Ignore all numbers except

$$2,\ 3,\ 5.$$

Eventually one of these three numbers will appear first.
Since each has equal probability,

$$P(2 \text{ appears first}) = \frac{1}{3}.$$

Hence

$$\boxed{\frac{1}{3}}$$

Therefore,

$$\boxed{(B)}$$

Alternative Proof

Let

$$p = P(2 \text{ occurs before } 3 \text{ or } 5).$$

On the first throw,

$$p = \frac{1}{6} + \frac{3}{6}p,$$

because

- Probability first throw is 2 is $\frac{1}{6}$.
- Probability first throw is 1,4 or 6 is $\frac{3}{6}$, after which the process restarts.

Therefore,

$$p - \frac{1}{2}p = \frac{1}{6},$$

which gives

$$\frac{1}{2}p = \frac{1}{6},$$

hence

$$p = \frac{1}{3}.$$

$$\boxed{\frac{1}{3}}$$

Question 8 (IIT JAM 2007)

Let E and F be two events such that

$$0 < P(E) < 1$$

and

$$P(E|F) + P(E|F^c) = 1.$$

Then

(A) E and F are mutually exclusive
(B) $P(E^c|F) + P(E^c|F^c) = 1$
(C) E and F are independent
(D) $P(E|F) + P(E^c|F^c) = 1$

Concept Used

Conditional Probability

$$P(E^c|F) = 1 - P(E|F)$$

Also,

$$P(E^c|F^c) = 1 - P(E|F^c).$$

Complement Rule:

$$P(A) + P(A^c) = 1.$$

Solution

Given

$$P(E|F) + P(E|F^c) = 1.$$

Now,

$$P(E^c|F) = 1 - P(E|F),$$

and

$$P(E^c|F^c) = 1 - P(E|F^c).$$

Adding,

$$P(E^c|F) + P(E^c|F^c) = 2 - \left[P(E|F) + P(E|F^c)\right].$$

Using the given condition,

$$= 2 - 1 = 1.$$

Hence,

$$\boxed{P(E^c|F) + P(E^c|F^c) = 1.}$$

Therefore,

$$\boxed{\textbf{Option (B)}}$$

Why Other Options are Incorrect

Option (A)

Nothing in the question implies

$$P(E \cap F) = 0.$$

Hence not necessarily mutually exclusive.

Option (C)
Independence requires

$$P(E|F) = P(E).$$

No such information is given.

Option (D)
Given

$$P(E|F) + P(E^c|F^c) = P(E|F) + 1 - P(E|F^c),$$

which is generally not equal to 1.
Hence false.

Exam Shortcut

Whenever you see

$$P(A|B) + P(A|B^c),$$

immediately think of the complement identity

$$P(A^c|B) = 1 - P(A|B).$$

This converts the expression in one line.

Answer: (B)

Question 9 (IIT JAM 2008)

One coin is selected at random from two coins.
The probability of obtaining head for one coin is

$$\frac{1}{3}$$

and for the other coin it is

$$\frac{1}{2}.$$

If the selected coin is tossed and the head shows up, find the probability that the selected coin is the fair coin.

Concept Used

Bayes' Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}.$$

where

$$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c).$$

Solution

Let

$$F = \{\text{fair coin is selected}\}$$

and

$$H = \{\text{head appears}\}.$$

Since one of the two coins is chosen randomly,

$$P(F) = \frac{1}{2}.$$

Also,

$$P(F^c) = \frac{1}{2}.$$

The probability of head for the fair coin is

$$P(H|F) = \frac{1}{2}.$$

The probability of head for the biased coin is

$$P(H|F^c) = \frac{1}{3}.$$

Using Bayes theorem,

$$P(F|H) = \frac{P(H|F)P(F)}{P(H)}.$$

First calculate

$$P(H) = P(H|F)P(F) + P(H|F^c)P(F^c).$$

Hence,

$$P(H) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}.$$

Therefore,

$$P(F|H) = \frac{\frac{1}{2} \cdot \frac{1}{2}}{\frac{5}{12}} = \frac{1}{4} \times \frac{12}{5} = \frac{3}{5}.$$

Hence,

$$\boxed{P(\text{Fair Coin}|\text{Head}) = \frac{3}{5}.}$$

Exam Shortcut

Whenever two objects are chosen randomly and an outcome is observed, immediately use

$$\boxed{\text{Bayes Theorem}}$$

Draw the probability tree first and then apply

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}.$$

$$\boxed{\text{Answer: } \frac{3}{5}}$$

Question 10 (IIT JAM 2008)

Let p denote the probability that the weather (either wet or dry) tomorrow will be the same as that of today.

If the weather is dry today, show that P_n, the probability that it will be dry n days later, satisfies

$$P_n = (2p-1)P_{n-1} + (1-p).$$

Hence, determine the value of P_{50} for

$$p = \frac{3}{4}.$$

Concept Used

This problem is based on a two-state Markov Chain.

States:

$$D = \text{Dry}, \qquad W = \text{Wet}$$

Transition probabilities are

$$P(D \to D) = p,$$

$$P(D \to W) = 1 - p,$$

$$P(W \to W) = p,$$

$$P(W \to D) = 1 - p.$$

Use the Law of Total Probability.

Solution

Let

$$P_n = P(\text{Dry on day } n).$$

On day $(n-1)$,
either the weather is Dry or Wet.
Hence,

$$P_n = P(D_n|D_{n-1})P_{n-1} + P(D_n|W_{n-1})(1 - P_{n-1}).$$

Substitute the transition probabilities:

$$P_n = pP_{n-1} + (1-p)(1 - P_{n-1}).$$

Expand:

$$P_n = pP_{n-1} + 1 - p - (1-p)P_{n-1}.$$

Collect like terms:

$$P_n = (p - (1-p))P_{n-1} + (1-p).$$

Therefore,

$$\boxed{P_n = (2p-1)P_{n-1} + (1-p).}$$

Hence proved.

For

$$p = \frac{3}{4},$$

the recurrence becomes

$$P_n = \frac{1}{2}P_{n-1} + \frac{1}{4}.$$

Initially,

$$P_0 = 1$$

because today is dry.
Subtract the steady-state value:
Let

$$Q_n = P_n - \frac{1}{2}.$$

Then

$$Q_n = \frac{1}{2} Q_{n-1}.$$

Hence,

$$Q_n = \left(\frac{1}{2}\right)^n Q_0.$$

Since

$$Q_0 = 1 - \frac{1}{2} = \frac{1}{2},$$

we obtain

$$Q_n = \left(\frac{1}{2}\right)^{n+1}.$$

Thus,

$$P_n = \frac{1}{2} + \left(\frac{1}{2}\right)^{n+1}.$$

Therefore,

$$P_{50} = \frac{1}{2} + \frac{1}{2^{51}}.$$

Hence,

$$\boxed{P_{50} = \frac{1}{2} + \frac{1}{2^{51}}.}$$

Shortcut

For recurrences of the form

$$x_n = a x_{n-1} + b,$$

the steady-state value is

$$x = \frac{b}{1-a}.$$

Write

$$y_n = x_n - x,$$

which gives

$$y_n = ay_{n-1}.$$

This converts the recurrence into a simple geometric progression.

Question 11 (IIT JAM 2009)

For detecting a disease, a test gives correct diagnosis with probability 0.99.

It is known that 1% of a population suffers from this disease.

If a randomly selected individual from this population tests positive, then the probability that the individual actually has the disease is

$$(A)\ 0.01 \qquad (B)\ 0.05 \qquad (C)\ 0.50 \qquad (D)\ 0.99$$

Concept Used

Bayes' Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}.$$

Solution

Let

$$D = \{\text{Person has disease}\},$$

$$T = \{\text{Test is positive}\}.$$

Given

$$P(D) = 0.01,$$

$$P(D^c) = 0.99.$$

Since the diagnosis is correct with probability

$$0.99,$$

we have

$$P(T|D) = 0.99,$$

and

$$P(T|D^c) = 0.01.$$

First calculate

$$P(T) = P(T|D)P(D) + P(T|D^c)P(D^c).$$

Therefore,

$$P(T) = 0.99(0.01) + 0.01(0.99) = 0.0198.$$

Now apply Bayes theorem:

$$P(D|T) = \frac{0.99 \times 0.01}{0.0198} = \frac{0.0099}{0.0198} = 0.5.$$

Hence,

$$\boxed{P(D|T) = 0.5.}$$

Therefore,

$$\boxed{\textbf{Option (C)}}$$

Interpretation

Although the test is 99% accurate, the disease is very rare (only 1% prevalence). Therefore, among all positive test results, many are false positives, reducing the probability that a positive test actually indicates the disease to only 50%.

Shortcut

Whenever a medical testing problem gives

- Disease prevalence,
- Test accuracy,
- Positive test result,

immediately apply Bayes' theorem.
Always compute

$$P(\text{Positive}) = P(\text{Positive}|D)P(D) + P(\text{Positive}|D^c)P(D^c).$$

Then substitute into Bayes' formula.

Answer: 0.5

Question 12 (IIT JAM 2009)

Let $U_1, U_2, \ldots, U_n$ be n urns such that urn U_k contains k white and k^2 black balls, where $k = 1, 2, \ldots, n$.

Suppose the probability of selecting urn U_k is proportional to $(k+1)$.
Find

(a) the probability that the drawn ball is black.

(b) the probability that urn U_n was selected given that the drawn ball is white.

Concept Used

- Probability proportional to a quantity
- Law of Total Probability
- Bayes' Theorem

Theory

If

$$P(U_k) \propto (k+1),$$

then

$$P(U_k) = \frac{k+1}{\sum_{i=1}^{n}(i+1)}.$$

Since

$$\sum_{i=1}^{n}(i+1) = \frac{n(n+3)}{2},$$

we obtain

$$\boxed{P(U_k) = \frac{2(k+1)}{n(n+3)}.}$$

—

Part (a)

Probability of drawing a black ball from urn U_k is

$$P(B|U_k) = \frac{k^2}{k+k^2} = \frac{k}{k+1}.$$

Using Total Probability,

$$P(B) = \sum_{k=1}^{n} P(B|U_k)P(U_k).$$

Substituting,

$$P(B) = \sum_{k=1}^{n} \frac{k}{k+1} \cdot \frac{2(k+1)}{n(n+3)}.$$

The factor $(k+1)$ cancels,

$$P(B) = \frac{2}{n(n+3)} \sum_{k=1}^{n} k.$$

Since

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2},$$

we get

$$P(B) = \frac{n+1}{n+3}.$$

Hence,

$$\boxed{P(B) = \frac{n+1}{n+3}.}$$

—

Part (b)

Probability of drawing a white ball from urn U_k

$$P(W|U_k) = \frac{k}{k+k^2} = \frac{1}{k+1}.$$

Hence,

$$P(W) = 1 - P(B) = \frac{2}{n+3}.$$

Now,

$$P(U_n) = \frac{2(n+1)}{n(n+3)}.$$

Using Bayes theorem,

$$P(U_n|W) = \frac{P(W|U_n)P(U_n)}{P(W)}.$$

Substitute,

$$= \frac{\frac{1}{n+1} \cdot \frac{2(n+1)}{n(n+3)}}{\frac{2}{n+3}}.$$

Simplifying,

$$= \frac{1}{n}.$$

Therefore,

$$\boxed{P(U_n|W) = \frac{1}{n}.}$$

Final Answers

$$\boxed{P(B) = \frac{n+1}{n+3}}$$

$$\boxed{P(U_n|W) = \frac{1}{n}}$$

Exam Shortcut

Whenever probabilities are proportional,
first normalize them.
Then use

$$P(A) = \sum P(A|B_i)P(B_i)$$

followed by Bayes theorem if required.

Question 13 (IIT JAM 2010)

Let E and F be two events with

$$P(E) > 0,$$

$$P(F|E) = 0.3,$$

and

$$P(E \cap F^c) = 0.2.$$

Then determine $P(E)$.

$$(A)\ \frac{1}{7} \qquad (B)\ \frac{2}{7} \qquad (C)\ \frac{4}{7} \qquad (D)\ \frac{5}{7}$$

Concept Used

Conditional Probability

$$P(F|E) = \frac{P(E \cap F)}{P(E)}.$$

Also,

$$P(E) = P(E \cap F) + P(E \cap F^c).$$

—

Solution

Given

$$P(F|E) = 0.3.$$

Therefore,

$$P(E \cap F) = 0.3P(E).$$

Also,

$$P(E \cap F^c) = 0.2.$$

Now,

$$P(E) = P(E \cap F) + P(E \cap F^c).$$

Substitute,

$$P(E) = 0.3P(E) + 0.2.$$

Hence,

$$0.7P(E) = 0.2.$$

Therefore,

$$P(E) = \frac{0.2}{0.7} = \frac{2}{7}.$$

Thus,

$$\boxed{P(E) = \frac{2}{7}.}$$

Hence,

$$\boxed{\textbf{Option (B)}}$$

—

Alternative Method

Since

$$P(F^c|E) = 1 - 0.3 = 0.7,$$

we have

$$P(E \cap F^c) = 0.7P(E).$$

Given

$$P(E \cap F^c) = 0.2,$$

therefore

$$0.7P(E) = 0.2,$$

which immediately gives

$$P(E) = \frac{2}{7}.$$

—

Exam Shortcut

If

$$P(A|B)$$

and

$$P(B \cap A^c)$$

are given,
use

$$P(B \cap A^c) = P(A^c|B)P(B).$$

This usually gives the answer in one step.

$$\boxed{\text{Answer: } \frac{2}{7}}$$

Question 14 (IIT JAM 2010)

Two coins with probabilities of heads u and v, respectively, are tossed independently.

If

$$P(\text{both tails}) = P(\text{both heads}),$$

then find $u + v$.

$$(A)\ \frac{1}{4} \qquad (B)\ \frac{1}{2} \qquad (C)\ \frac{3}{4} \qquad (D)\ 1$$

Concept Used

Independent Events

For independent events,

$$P(A \cap B) = P(A)P(B).$$

Solution

Probability that both coins show heads is

$$P(HH) = uv.$$

Probability that both coins show tails is

$$P(TT) = (1-u)(1-v).$$

Given

$$uv = (1-u)(1-v).$$

Expanding,

$$uv = 1 - u - v + uv.$$

Cancelling uv from both sides,

$$u + v = 1.$$

Hence,

$$\boxed{u + v = 1.}$$

Therefore,

$$\boxed{\textbf{Option (D)}}$$

Shortcut

Whenever products appear on both sides,
expand one side and cancel common terms.

$$\boxed{u + v = 1}$$

Question 15 (IIT JAM 2011)

A fair die is rolled three times.

Find the conditional probability that the face 6 appears exactly once, given that it appeared at least once.

Concept Used

Conditional Probability

$$P(A|B) = \frac{P(A)}{P(B)}.$$

Binomial Distribution

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}.$$

Solution

Let

$$A = \{\text{Exactly one six}\},$$

$$B = \{\text{At least one six}\}.$$

Clearly,

$$A \subseteq B.$$

Therefore,

$$P(A|B) = \frac{P(A)}{P(B)}.$$

Probability of exactly one six is

$$P(A) = \binom{3}{1}\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^2.$$

Probability of at least one six is

$$P(B) = 1 - \left(\frac{5}{6}\right)^3.$$

Hence,

$$P(A|B) = \frac{3\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^2}{1-\left(\frac{5}{6}\right)^3}.$$

Therefore,

$$\boxed{\frac{3\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^2}{1-\left(\frac{5}{6}\right)^3}}$$

Hence,

$$\boxed{\textbf{Option (C)}}$$

Shortcut

Whenever "given at least one" appears,
use

$$P(\text{at least one}) = 1 - P(\text{none}).$$

Question 16 (IIT JAM 2011)

A university has

30% Asian,

40% American,

20% European,

10% African students.

Among them,

40% Asians are girls,

50% Americans are girls,

60% Europeans are girls,

20% Africans are girls.

Find the probability that a randomly chosen girl is Asian.

Concept Used

Bayes' Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}.$$

Solution

Let

$$A = \{\text{Asian}\},$$

$$G = \{\text{Girl}\}.$$

Given

$$P(A) = 0.30,$$

$$P(Am) = 0.40,$$

$$P(E) = 0.20,$$

$$P(Af) = 0.10.$$

Also,

$$P(G|A) = 0.40,$$

$$P(G|Am) = 0.50,$$

$$P(G|E) = 0.60,$$

$$P(G|Af) = 0.20.$$

First compute

$$P(G) = 0.30(0.40) + 0.40(0.50) + 0.20(0.60) + 0.10(0.20).$$

Therefore,

$$P(G) = 0.12 + 0.20 + 0.12 + 0.02 = 0.46.$$

Using Bayes theorem,

$$P(A|G) = \frac{0.40 \times 0.30}{0.46} = \frac{0.12}{0.46} = \frac{6}{23}.$$

Hence,

$$\boxed{P(\text{Asian}|\text{Girl}) = \frac{6}{23}.}$$

Shortcut

Construct a probability table first.

Then compute

$$P(\text{Girl}) = \sum P(\text{Girl}|\text{Group})P(\text{Group}),$$

and finally apply Bayes theorem.

$$\boxed{\frac{6}{23}}$$

Question 17 (IIT JAM 2011)

Let A_1, A_2 and A_3 be pairwise independent events with

$$P(A_i) = \frac{1}{2}, \qquad i = 1, 2, 3.$$

Suppose that A_3 and $A_1 \cup A_2$ are independent. Find

$$P(A_1 \cap A_2 \cap A_3).$$

Concept Used

- Pairwise Independence
- Independence of Events
- Inclusion-Exclusion Principle

Theory

If two events E and F are independent,

$$P(E \cap F) = P(E)P(F).$$

Also,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Solution

Since the events are pairwise independent,

$$P(A_1 \cap A_2) = P(A_1)P(A_2) = \frac{1}{4}.$$

Therefore,

$$P(A_1 \cup A_2) = \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}.$$

Now,

$$A_3 \text{ and } (A_1 \cup A_2) \text{ are independent.}$$

Hence,

$$P\left(A_3 \cap (A_1 \cup A_2)\right) = P(A_3)P(A_1 \cup A_2).$$

Therefore,

$$= \frac{1}{2} \times \frac{3}{4} = \frac{3}{8}.$$

Now,

$$A_3 \cap (A_1 \cup A_2) = (A_1 \cap A_3) \cup (A_2 \cap A_3).$$

Applying Inclusion-Exclusion,

$$P = P(A_1 \cap A_3) + P(A_2 \cap A_3) - P(A_1 \cap A_2 \cap A_3).$$

Since pairwise independent,

$$P(A_1 \cap A_3) = \frac{1}{4},$$

$$P(A_2 \cap A_3) = \frac{1}{4}.$$

Hence,

$$\frac{3}{8} = \frac{1}{4} + \frac{1}{4} - P(A_1 \cap A_2 \cap A_3).$$

Thus,

$$P(A_1 \cap A_2 \cap A_3) = \frac{1}{2} - \frac{3}{8} = \frac{1}{8}.$$

$$\boxed{P(A_1 \cap A_2 \cap A_3) = \frac{1}{8}.}$$

Exam Shortcut

Always compute

$$P(A_1 \cup A_2)$$

first and then use

$$A_3 \cap (A_1 \cup A_2) = (A_1 \cap A_3) \cup (A_2 \cap A_3).$$

This avoids lengthy calculations.

$$\boxed{\text{Answer: } \frac{1}{8}}$$

Question 18 (IIT JAM 2012)

A person makes repeated attempts to destroy a target.

Attempts are independent.

The probability of destroying the target in any attempt is

$$0.8.$$

Given that the person fails to destroy the target in the first five attempts, find the probability that the target is destroyed in the eighth attempt.

$$(A)\ 0.128 \qquad (B)\ 0.032 \qquad (C)\ 0.160 \qquad (D)\ 0.064$$

Concept Used

- Conditional Probability
- Independent Trials
- Geometric Distribution

Theory

Since the trials are independent,
previous failures do not affect future attempts.
If success probability is

$$p,$$

then

$$P(\text{first success on } k^{th} \text{ trial}) = (1-p)^{k-1}p.$$

Solution

Let

$$F = \{\text{Failure in first five attempts}\},$$

and

$$E = \{\text{Target destroyed in 8th attempt}\}.$$

Since the first five failures are already known,
the person must

- fail in attempts 6 and 7,
- succeed in attempt 8.

Because attempts are independent,

$$P(E|F) = (0.2)^2(0.8).$$

Therefore,

$$= 0.04 \times 0.8 = 0.032.$$

Hence,

$$\boxed{P(E|F) = 0.032.}$$

Therefore,

$$\boxed{\textbf{Option (B)}}$$

Alternative Method

Using conditional probability,

$$P(E|F) = \frac{P(F \cap E)}{P(F)}.$$

Now,

$$P(F \cap E) = (0.2)^7(0.8),$$

and

$$P(F) = (0.2)^5.$$

Hence,

$$P(E|F) = \frac{(0.2)^7(0.8)}{(0.2)^5} = (0.2)^2(0.8) = 0.032.$$

Exam Shortcut

Whenever the trials are independent,
ignore everything before the given condition.
Start counting from the conditioned trial.
Only multiply the probabilities required after that point.

$$\boxed{\text{Answer: } 0.032}$$

Question 19 (IIT JAM 2012)

A four-digit number is chosen at random. The probability that there are exactly two zeros in that number is

$$(A)\ 0.73 \qquad (B)\ 0.973 \qquad (C)\ 0.027 \qquad (D)\ 0.27$$

Concept Used

- Counting Principle
- Classical Probability
- Combinations

Theory

A four-digit number ranges from

$$1000 \text{ to } 9999.$$

Hence,

$$\boxed{\text{Total four-digit numbers} = 9000.}$$

Probability is

$$P(E) = \frac{\text{Favourable Outcomes}}{\text{Total Outcomes}}.$$

Solution

A four-digit number has the form

$$d_1 d_2 d_3 d_4,$$

where

$$d_1 \neq 0.$$

Exactly two digits must be zero.
Since the first digit cannot be zero, the two zeros must occur among

$$d_2, d_3, d_4.$$

Choose two positions for zeros:

$$\binom{3}{2} = 3.$$

The remaining position among

$$d_2, d_3, d_4$$

can be filled in

$$9$$

ways (digits 1 to 9).
The first digit also has

$$9$$

choices.
Hence,

$$\text{Favourable numbers} = 9 \times 3 \times 9 = 243.$$

Therefore,

$$P = \frac{243}{9000} = 0.027.$$

Hence,

$$\boxed{P = 0.027.}$$

Therefore,

$$\boxed{\textbf{Option (C)}}$$

Exam Shortcut

For four-digit numbers,

the first digit can never be zero.

Hence the two zeros must come only from the last three positions.

Immediately write

$$9 \times \binom{3}{2} \times 9.$$

$$\boxed{\text{Answer: } 0.027}$$

Question 20 (IIT JAM 2013)

Let E and F be two events such that

$$P(E) = 0.7,$$

$$P(F) = 0.4,$$

and

$$P(E \cap F^c) = 0.4.$$

Find

$$P(F \mid E \cup F).$$

$$(A)\ \frac{1}{2} \qquad (B)\ \frac{1}{3} \qquad (C)\ \frac{1}{4} \qquad (D)\ \frac{1}{5}$$

Concept Used

- Conditional Probability
- Inclusion-Exclusion Principle

Theory

Conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

Also,

$$P(E \cup F) = P(E) + P(F) - P(E \cap F).$$

Solution

Given

$$P(E) = 0.7,$$

$$P(E \cap F^c) = 0.4.$$

Since

$$P(E) = P(E \cap F) + P(E \cap F^c),$$

we get

$$P(E \cap F) = 0.7 - 0.4 = 0.3.$$

Now,

$$P(E \cup F) = 0.7 + 0.4 - 0.3 = 0.8.$$

Observe that

$$F \subseteq (E \cup F),$$

therefore,

$$F \cap (E \cup F) = F.$$

Hence,

$$P(F|E \cup F) = \frac{P(F)}{P(E \cup F)} = \frac{0.4}{0.8} = \frac{1}{2}.$$

Thus,

$$\boxed{P(F|E \cup F) = \frac{1}{2}.}$$

Therefore,

$$\boxed{\textbf{Option (A)}}$$

Alternative Method

Using

$$P(E \cap F) = 0.3,$$

write

$$P(F|E \cup F) = \frac{P(F \cap (E \cup F))}{P(E \cup F)} = \frac{0.4}{0.8} = \frac{1}{2}.$$

Exam Shortcut

Whenever the condition is

$$E \cup F,$$

remember

$$F \subseteq (E \cup F),$$

so the numerator immediately becomes

$$P(F).$$

Only calculate

$$P(E \cup F)$$

using Inclusion-Exclusion.

$$\boxed{\text{Answer: } \frac{1}{2}}$$

Question 21 (IIT JAM 2013)

Ten percent of the bolts produced in a factory are defective. They are randomly packed into boxes such that each box contains 3 bolts.

Four of these boxes are bought by a customer.

Find the probability that none of the four boxes contains any defective bolt.

Concept Used

- Independence of Bolts
- Multiplication Rule of Probability
- Binomial Probability

Theory

Let

$$P(D) = 0.1$$

be the probability that a bolt is defective.

Therefore,

$$P(G) = 0.9$$

is the probability that a bolt is non-defective.
Since the bolts are packed randomly and independently,

$$P(\text{All bolts in a box are good}) = (0.9)^3.$$

Similarly, if several boxes are selected independently,

$$P(\text{All boxes contain only good bolts}) = \left[(0.9)^3\right]^n,$$

where n is the number of boxes.

Solution

Each box contains

$$3$$

bolts.
Probability that one bolt is non-defective is

$$0.9.$$

Hence,

$$P(\text{one box has no defective bolt}) = (0.9)^3.$$

The customer purchases

$$4$$

boxes.
Therefore,

$$P(\text{all four boxes have no defective bolt}) = \left((0.9)^3\right)^4.$$

Using the law of exponents,

$$= (0.9)^{12}.$$

Hence,

$$\boxed{P = (0.9)^{12}.}$$

Numerically,

$$(0.9)^{12} = 0.28243 \approx 0.282.$$

Thus,

$$\boxed{P = (0.9)^{12} \approx 0.282.}$$

Alternative Method

The customer receives

$$4 \times 3 = 12$$

bolts in total.

The event "none of the four boxes contains a defective bolt" is equivalent to saying

$$\text{all 12 bolts are non-defective.}$$

Hence,

$$P = (0.9)^{12}.$$

Exam Shortcut

Instead of considering the boxes one by one,

observe that

$$4 \text{ boxes } \times 3 \text{ bolts} = 12 \text{ bolts}.$$

Therefore,

$$P(\text{all good}) = (0.9)^{12}.$$

No lengthy calculation is required.

$$\boxed{\text{Final Answer } = (0.9)^{12} \approx 0.282}$$

Question 22 (IIT JAM 2013)

Ram rolls a pair of fair dice.

If the sum of the numbers on the upper faces is 5, 6, 10, 11 or 12, then Ram wins a gold coin.

Otherwise, he rolls the pair of dice once again and wins a silver coin if the sum obtained in the second throw is the same as the sum obtained in the first throw.

Find the probability that Ram wins either a gold coin or a silver coin.

Concept Used

- Law of Total Probability
- Mutually Exclusive Events
- Probability Distribution of the Sum of Two Dice

Theory

For two fair dice,

$$\text{Total outcomes} = 36.$$

The distribution of the sum is

Sum	2	3	4	5	6	7	8	9	10	11	12
Ways	1	2	3	4	5	6	5	4	3	2	1

Hence,

$$P(\text{Sum} = s) = \frac{\text{Number of ways}}{36}.$$

Solution

Let

$$G = \{\text{Wins Gold Coin}\},$$

$$S = \{\text{Wins Silver Coin}\}.$$

These events are mutually exclusive because the second throw is made only if Ram does not win the gold coin.

Hence,

$$P(G \cup S) = P(G) + P(S).$$

Step 1: Probability of Winning Gold Coin

Gold is won when the first sum is

$$5, 6, 10, 11, 12.$$

Number of favourable outcomes

$$4 + 5 + 3 + 2 + 1 = 15.$$

Therefore,

$$P(G) = \frac{15}{36} = \frac{5}{12}.$$

Step 2: Probability of Winning Silver Coin

Silver is possible only when the first sum is

$$2, 3, 4, 7, 8, 9,$$

whose frequencies are

$$1, 2, 3, 6, 5, 4.$$

If the first sum equals s, then the probability that the second throw also has sum s is

$$\frac{\text{Frequency of } s}{36}.$$

Hence,

$$P(S) = \sum P(\text{First sum} = s)P(\text{Second sum} = s).$$

Therefore,

$$P(S) = \frac{1^2 + 2^2 + 3^2 + 6^2 + 5^2 + 4^2}{36^2}.$$

Now,

$$1 + 4 + 9 + 36 + 25 + 16 = 91.$$

Hence,

$$P(S) = \frac{91}{1296}.$$

Step 3: Total Probability

Therefore,

$$P(G \cup S) = \frac{15}{36} + \frac{91}{1296}.$$

Since

$$\frac{15}{36} = \frac{540}{1296},$$

we obtain

$$P(G \cup S) = \frac{540 + 91}{1296} = \frac{631}{1296}.$$

Hence,

$$\boxed{P(\text{Gold or Silver}) = \frac{631}{1296}.}$$

Numerically,

$$\boxed{P \approx 0.4869.}$$

Alternative Method

Use the Law of Total Probability.

For each non-gold sum,
multiply

$$P(\text{first sum}) \times P(\text{same sum again}),$$

and then add the gold probability.

Exam Shortcut

Remember the frequency table of the sum of two dice.

Gold sums:

$$5, 6, 10, 11, 12$$

have frequencies

$$4, 5, 3, 2, 1.$$

Remaining sums have frequencies

$$1, 2, 3, 6, 5, 4.$$

Silver probability is simply

$$\frac{\sum f_i^2}{36^2}.$$

Thus,

$$P(S) = \frac{91}{1296}.$$

Finally,

$$P(G \cup S) = \frac{15}{36} + \frac{91}{1296} = \boxed{\frac{631}{1296}}.$$

Question 23 (IIT JAM 2014)

A system consisting of n components functions if, and only if, at least one of the n components functions.

Suppose all the components function independently, each with probability

$$\frac{3}{4}.$$

If the probability that the system functions is

$$\frac{63}{64},$$

find the value of n.

$$(A)\ 2 \qquad (B)\ 4 \qquad (C)\ 3 \qquad (D)\ 5$$

Concept Used

- Complement Rule
- Independent Events
- Reliability of a Parallel System

Theory

The system functions if at least one component functions.

Hence,

$$P(\text{System Works}) = 1 - P(\text{All Components Fail}).$$

Since every component works with probability

$$\frac{3}{4},$$

the probability that one component fails is

$$\frac{1}{4}.$$

Because the components are independent,

$$P(\text{All Fail}) = \left(\frac{1}{4}\right)^n.$$

Solution

Given,

$$P(\text{System Works}) = \frac{63}{64}.$$

Therefore,

$$1 - \left(\frac{1}{4}\right)^n = \frac{63}{64}.$$

Hence,

$$\left(\frac{1}{4}\right)^n = \frac{1}{64}.$$

Now,

$$64 = 4^3.$$

Thus,

$$\left(\frac{1}{4}\right)^n = \left(\frac{1}{4}\right)^3.$$

Therefore,

$$\boxed{n = 3.}$$

Hence,

$$\boxed{\textbf{Option (C)}}$$

Shortcut

For a parallel system,

$$P(\text{Working}) = 1 - (q)^n,$$

where

$$q = P(\text{Failure of one component}).$$

Simply equate powers.

$$\boxed{n = 3}$$

Question 24 (IIT JAM 2014)

Four persons P, Q, R and S take turns (in the order

$$P, Q, R, S, P, Q, R, S, \ldots$$

)
to roll a fair die.
The first person to obtain a six wins.
Find the probability that S wins.

$$(A)\ \frac{125}{671} \qquad (B)\ \frac{125}{571} \qquad (C)\ \frac{125}{471} \qquad (D)\ \frac{125}{371}$$

Concept Used

- Geometric Series
- Independent Trials
- Infinite Probability Series

Theory

Each player gets one chance in every cycle of four throws.

The probability of not getting a six in one throw is

$$\frac{5}{6}.$$

Therefore,
the probability that nobody wins in one complete cycle is

$$\left(\frac{5}{6}\right)^4.$$

Solution

For S to win in the first cycle,

- P fails,
- Q fails,
- R fails,
- S gets a six.

Hence,

$$P(\text{S wins in first cycle}) = \left(\frac{5}{6}\right)^3 \left(\frac{1}{6}\right) = \frac{125}{1296}.$$

If nobody wins in the first cycle,
the game starts again.
Probability that nobody wins in one cycle is

$$r = \left(\frac{5}{6}\right)^4 = \frac{625}{1296}.$$

Hence,

$$P(S) = \frac{125}{1296} + \frac{625}{1296} \cdot \frac{125}{1296} + \left(\frac{625}{1296}\right)^2 \frac{125}{1296} + \cdots$$

This is an infinite geometric series.
Therefore,

$$P(S) = \frac{\frac{125}{1296}}{1 - \frac{625}{1296}}.$$

Now,

$$1296 - 625 = 671.$$

Hence,

$$P(S) = \frac{125}{671}.$$

Therefore,

$$\boxed{P(S) = \frac{125}{671}.}$$

Thus,

$$\boxed{\textbf{Option (A)}}$$

Alternative Method

Using geometric series,

$$P = a + ar + ar^2 + \cdots = \frac{a}{1 - r},$$

where

$$a = \left(\frac{5}{6}\right)^3 \frac{1}{6},$$

and

$$r = \left(\frac{5}{6}\right)^4.$$

Shortcut

Whenever players take turns repeatedly,
find

$$\text{(Probability of winning in first cycle)}$$

and multiply it by

$$\frac{1}{1-r},$$

where

$$r = P(\text{No winner in one cycle}).$$

$$\boxed{\text{Answer } = \frac{125}{671}}$$

Question 25 (IIT JAM 2014)

There are two urns U_1 and U_2.

Urn U_1 contains four white and four black balls, while urn U_2 is initially empty.

Four balls are selected at random from U_1 and transferred to U_2.

One ball is then drawn at random from U_2.

Find the probability that the ball drawn from U_2 is white.

$$(A)\ \frac{1}{3} \qquad (B)\ \frac{1}{2} \qquad (C)\ \frac{2}{3} \qquad (D)\ \frac{3}{4}$$

Concept Used

- Law of Total Probability
- Hypergeometric Distribution
- Linearity of Expectation (Alternative Method)

Theory

Suppose exactly k white balls are transferred.

Then

$$P(\text{White from } U_2 \mid k \text{ whites}) = \frac{k}{4}.$$

Hence,

$$P(\text{White}) = \sum_{k=0}^{4} P(\text{White} \mid k) P(k).$$

The number of white balls transferred follows a Hypergeometric distribution.

Solution

Let

$$X = \text{Number of white balls transferred to } U_2.$$

Since four balls are selected from

$$4 \text{ white}, \qquad 4 \text{ black},$$

the probability that exactly k white balls are transferred is

$$P(X = k) = \frac{\binom{4}{k}\binom{4}{4-k}}{\binom{8}{4}}, \qquad k = 0, 1, 2, 3, 4.$$

Now,

$$P(\text{White} \mid X = k) = \frac{k}{4}.$$

Therefore,

$$P(\text{White}) = \sum_{k=0}^{4} \frac{k}{4} \cdot \frac{\binom{4}{k}\binom{4}{4-k}}{\binom{8}{4}}.$$

Instead of evaluating the entire sum, use the expected value.

Since

$$E(X) = 4 \times \frac{4}{8} = 2,$$

we have

$$P(\text{White}) = \frac{E(X)}{4} = \frac{2}{4} = \frac{1}{2}.$$

Hence,

$$\boxed{P(\text{White}) = \frac{1}{2}.}$$

Therefore,

$$\boxed{\textbf{Option (B)}}$$

Alternative Method

Using the symmetry of the urn,
initially,

$$\frac{1}{2}$$

of the balls are white.

Randomly selecting four balls does not change the expected proportion of white balls in the transferred sample.

Hence,

$$P(\text{White}) = \frac{1}{2}.$$

Exam Shortcut

Whenever a random sample is transferred,
remember

$$P(\text{White}) = \frac{E(\text{White balls transferred})}{\text{Sample size}}.$$

Since

$$E(X) = n \cdot \frac{W}{W+B},$$

here

$$E(X) = 4 \times \frac{4}{8} = 2.$$

Therefore,

$$P(\text{White}) = \frac{2}{4} = \boxed{\frac{1}{2}.}$$

Question 26 (IIT JAM 2014)

Suppose that five distinct balls are distributed at random into three distinct boxes in such a way that each of the five balls can be placed into any one of the three boxes.

Find

(a) the probability that exactly one box is empty.

(b) the probability that all three boxes are occupied.

Concept Used

- Counting Principle
- Inclusion–Exclusion Principle
- Classical Probability

Theory

Each of the five balls has three choices.

Hence,

$$\boxed{\text{Total possible distributions} = 3^5 = 243.}$$

The required probability is

$$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total outcomes}}.$$

Part (a): Exactly One Box is Empty

Exactly one of the three boxes is empty.

Step 1: Choose the Empty Box

There are

$$\binom{3}{1} = 3$$

ways.

Step 2: Distribute Balls

The five balls must now be distributed into the remaining two boxes so that neither box is empty.

Each ball has two choices,

$$2^5$$

assignments.

Subtract the two cases where all balls go to only one box.

Hence,

$$2^5 - 2 = 30.$$

Therefore,

$$\text{Favourable outcomes} = 3 \times 30 = 90.$$

Thus,

$$P(\text{Exactly one empty}) = \frac{90}{243} = \boxed{\frac{10}{27}}.$$

Part (b): All Boxes Occupied

Use the complement or Inclusion–Exclusion Principle.

Total outcomes

$$3^5 = 243.$$

Cases with at least one empty box:

$$3(2^5) - 3(1^5) = 96 - 3 = 93.$$

Hence,

$$\text{All occupied} = 243 - 93 = 150.$$

Therefore,

$$P(\text{All occupied}) = \frac{150}{243} = \boxed{\frac{50}{81}}.$$

Alternative Method

Using Inclusion–Exclusion,

$$N = 3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5.$$

Hence,

$$N = 243 - 96 + 3 = 150.$$

Therefore,

$$P = \frac{150}{243} = \frac{50}{81}.$$

Exam Shortcut

Remember the formulas:

Exactly one box empty

$$= \binom{3}{1}(2^5 - 2).$$

All boxes occupied

$$= 3^5 - \binom{3}{1}2^5 + \binom{3}{2}.$$

These are standard occupancy results for distinct balls and distinct boxes.

$$\boxed{P(\text{Exactly one empty}) = \frac{10}{27}}$$

$$\boxed{P(\text{All occupied}) = \frac{50}{81}}$$